



MANAV RACHNA UNIVERSITY
(NAAC A++ Accredited)



LARSEN & TOUBRO

Engineering the Future

DESIGN THINKING BOARD

A DESIGN THINKING APPROACH TO SOLVING
REAL-WORLD PROBLEMS

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1. Introduction

This document describes the process of creating a Design Thinking Board for a Smart Water Bottle. The objective was to identify a real-world problem related to hydration and develop an innovative solution using the Design Thinking methodology.

Lucid Chart Link

https://lucid.app/lucidchart/2b5800dc-debd-46d5-a0e1-0233a8edcf1b/edit?viewport_loc=-561%2C-407%2C3207%2C1542%2C0_0&invitationId=inv_5c6f2246-d439-4086-9f31-17a70f91bf79

2. Product Selection

The selected product is a **Smart Water Bottle**, designed to help users monitor and improve their daily water intake through smart features such as reminders, tracking, and mobile connectivity.

3. Problem Identification

Many individuals fail to maintain proper hydration due to:

- Busy lifestyles
- Lack of awareness
- Absence of tracking systems
- Ineffective reminder methods

This leads to health issues and reduced productivity

4. Design Thinking Process

a) Empathize

We analyzed user behavior and identified key challenges:

- Users forget to drink water
- Hydration is not prioritized
- No easy tracking mechanism exists
- Existing reminders are often ignored

b) Define

Problem Statement:

People fail to maintain proper hydration due to lack of awareness, tracking tools, and motivation.

c) Ideate

Possible solutions were brainstormed:

- Smart bottle with automated reminders
- Mobile application for tracking water intake
- LED indicators for alerts
- Gamification features (goals and rewards)
- Temperature monitoring system

d) Prototype

A basic solution model was designed including:

- Bottle with embedded sensors
- Mobile app interface
- Bluetooth connectivity
- Rechargeable battery system
- Initial working prototype

e) Test

The prototype was evaluated through:

- Testing reminder effectiveness
- Checking app usability
- Collecting user feedback
- Identifying and fixing bugs
- Improving overall functionality

5. Tools Used

- Lucidchart was used to create the Design Thinking Board
- Sticky notes and structured columns were used to represent each stage clearly

6. Conclusion

The Design Thinking approach helped in systematically identifying user problems and developing a practical solution. The Smart Water Bottle concept demonstrates how technology can be used to improve daily habits and promote better health

THANK YOU